

COMPANION FOR CHAPTER 10

OF

**FUNDAMENTALS
OF PROBABILITY**

WITH STOCHASTIC PROCESSES

FOURTH EDITION

SAEED GHAHRAMANI

Western New England University
Springfield, Massachusetts, USA



CRC Press

Taylor & Francis Group

Boca Raton London New York

CRC Press is an imprint of the
Taylor & Francis Group, an **informa** business
A CHAPMAN & HALL BOOK

Contents

► **10 More Expectations and Variances** **3**

10A	Additional Examples	3
10B	Pattern Appearance	16
10C	More on Memoryless Property of Exponential Random Variables	20
10D	Conditioning on More than One Random Variable	25
	Answers to Exercises of Section 10B	31

Chapter 10

More Expectations and Variances

10A ADDITIONAL EXAMPLES

Example 10a Three points A , B , and C are randomly placed on a circle. What is the probability that they lie on a semicircle?

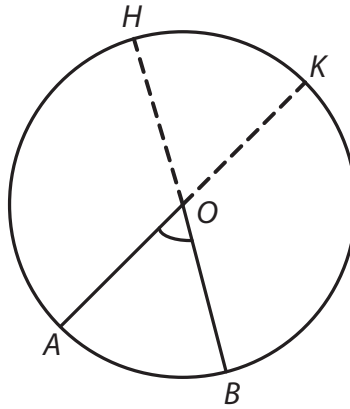


Figure 10c The circle of Example 10a.

Solution: Let O be the center of the circle. After A and B are placed, draw the diameter through A intersecting the circle at K (see Figure 10c). Also draw the diameter through B intersecting the circle at H . Let the measure of the acute angle AOB be X (note that the probability is 0 that the angle AOB is straight). The random variable X is uniform over the interval $(0, \pi)$. So f , the probability density function of X , is given by

$$f(x) = \begin{cases} 1/\pi & 0 < x < \pi \\ 0 & \text{otherwise.} \end{cases}$$

Let E be the event that C is placed on the arc HAK . Clearly, A , B , and C lie on a

semicircle if and only if E occurs. Thus the desired probability is

$$\begin{aligned} P(E) &= \int_0^\pi P(E \mid X = x) f(x) dx = \int_0^\pi P(E \mid X = x) \frac{1}{\pi} dx \\ &= \int_0^\pi \frac{2\pi - x}{2\pi} \cdot \frac{1}{\pi} dx = \frac{3}{4}. \end{aligned}$$

Remark: Note that A , B and C lie on a semicircle if and only if the triangle ABC is obtuse. This example implies the fact that 75% of “all triangles” are obtuse. That is, if from the class of all triangles a triangle is selected at random, then the probability is $3/4$ that it is obtuse. (See Exercise 8, Section 7.1 of the book.) ♦

Example 10b The lifetimes of batteries (in hours) manufactured by a certain company are identically distributed with distribution and probability density functions F and f , respectively.

- (a) In terms of F , f , and s , find the expected value of the lifetime of an s -hour-old battery.
- (b) Suppose that the lifetimes of the batteries are exponential random variables with identical parameter λ . Find the expected value of the lifetime of a s -hour-old device, and the expected value of the lifetime of a battery, which is not yet s hours old.

Solution: (a) Let X be the lifetime of the s -hour-old device. We want to calculate $E(X \mid X > s)$. Let

$$F_{X|X>s}(t) = P(X \leq t \mid X > s),$$

and $f_{X|X>s}(t) = F'_{X|X>s}(t)$. Then

$$E(X \mid X > s) = \int_0^\infty t f_{X|X>s}(t) dt.$$

Now

$$\begin{aligned} F_{X|X>s}(t) &= P(X \leq t \mid X > s) = \frac{P(X \leq t, X > s)}{P(X > s)} \\ &= \begin{cases} 0 & \text{if } t \leq s \\ \frac{P(s < X \leq t)}{P(X > s)} & \text{if } t > s. \end{cases} \end{aligned}$$

Therefore,

$$F_{X|X>s}(t) = \begin{cases} 0 & \text{if } t \leq s \\ \frac{F(t) - F(s)}{1 - F(s)} & \text{if } t > s. \end{cases}$$

Differentiating $F_{X|X>s}(t)$ with respect to t , we obtain

$$f_{X|X>s}(t) = \begin{cases} 0 & \text{if } t \leq s \\ \frac{f(t)}{1 - F(s)} & \text{if } t > s. \end{cases}$$

This yields

$$E(X | X > s) = \int_0^\infty t f_{X|X>s}(t) dt = \frac{1}{1 - F(s)} \int_s^\infty t f(t) dt.$$

(b) Given that the battery is s hours old, by part (a), its probability density function, $f_{X|X>s}$, is given by

$$f_{X|X>s}(t) = \begin{cases} 0 & \text{if } t \leq s \\ \lambda e^{-\lambda(t-s)} & \text{if } t > s, \end{cases}$$

and

$$E(X | X > s) = e^{\lambda s} \int_s^\infty \lambda t e^{-\lambda t} dt = s + \frac{1}{\lambda},$$

a result expected because of the memoryless property of exponential random variables. Now, by Theorem 10.10 of the book,

$$E(X) = E(X | X < s)P(X < s) + E(X | X > s)P(X > s),$$

which implies that

$$\frac{1}{\lambda} = E(X | X < s) \cdot (1 - e^{-\lambda s}) + \left(s + \frac{1}{\lambda}\right) \cdot e^{-\lambda s}.$$

Solving this equation for $E(X | X < s)$, we have

$$E(X | X < s) = \frac{1}{\lambda} - \frac{s e^{-\lambda s}}{1 - e^{-\lambda s}}.$$

Note that, in this example, we have established another important property of exponential random variables:

Let X be an exponential random variable with parameter λ . For $s > 0$,

$$E(X | X > s) = s + \frac{1}{\lambda},$$

$$E(X | X < s) = \frac{1}{\lambda} - \frac{s e^{-\lambda s}}{1 - e^{-\lambda s}}. \quad \blacklozenge$$

Example 10c A spice company distributes cinnamon in one-pound bags. Suppose that the Food and Drug Administration (FDA) considers more than 500 insect fragments in one bag excessive and hence unacceptable. To meet the standards of the FDA, the quality control division of the company begins inspecting the bags of the cinnamon at time $t = 0$ according to the following scheme. It inspects each bag with probability α until it encounters an unacceptable bag. At that point, the division inspects every single bag until it finds m consecutive acceptable bags. When this happens, the division has completed one inspection cycle. It then resumes its normal inspection process. Inspected bags that are found with excessive numbers of insect fragments are sent back for further cleaning. Let p be the probability that a bag is acceptable, independent of the number of insect fragments in other bags. Find the expected value of the number of bags inspected in one inspection cycle.

Solution: Let X be the number of bags inspected until an unacceptable bag is found. Let K_n be the number of consequent bags inspected until n consecutive acceptable bags are found. The number of bags inspected in one inspection cycle is $X + K_m$. We are interested in $E(X + K_m) = E(X) + E(K_m)$. Clearly, X is a geometric random variable with parameter $\alpha(1 - p)$. So $E(X) = 1/[\alpha(1 - p)]$. To find $E(K_m)$, note that $\forall n$,

$$E(K_n) = E[E(K_n | K_{n-1})].$$

Now

$$\begin{aligned} E(K_n | K_{n-1} = i) &= (i + 1)p + [i + 1 + E(K_n)](1 - p) \\ &= (i + 1) + (1 - p)E(K_n). \end{aligned} \tag{1}$$

To derive this relation, we noted the following. It took i inspections to find $n - 1$ consecutive acceptable bags. If the next bag inspected is also acceptable, we have the n consecutive acceptable bags required in $i + 1$ inspections. This occurs with probability p . However, if the next bag inspected is unacceptable, then, on the average, we need an additional $E(K_n)$ inspections [a total of $i + 1 + E(K_n)$ inspections] until we get n consecutive acceptable bags of cinnamon. This happens with probability $1 - p$.

From (1), we have

$$E(K_n | K_{n-1}) = (K_{n-1} + 1) + (1 - p)E(K_n).$$

Finding the expected values of both sides of this relation gives

$$E(K_n) = E(K_{n-1}) + 1 + (1 - p)E(K_n).$$

Solving for $E(K_n)$, we obtain

$$E(K_n) = \frac{1}{p} + \frac{E(K_{n-1})}{p}.$$

Noting that $E(K_1) = 1/p$ and solving recursively, we find that

$$E(K_n) = \frac{1}{p} + \frac{1}{p^2} + \cdots + \frac{1}{p^n}.$$

Therefore, the desired quantity is

$$\begin{aligned}
 E(X + K_m) &= E(X) + E(K_m) \\
 &= \frac{1}{\alpha(1-p)} + \frac{1}{p} \left(1 + \frac{1}{p} + \cdots + \frac{1}{p^{m-1}} \right) \\
 &= \frac{1}{\alpha(1-p)} + \frac{1}{p} \cdot \frac{\left(\frac{1}{p}\right)^m - 1}{\frac{1}{p} - 1} = \frac{(1-\alpha)p^m + \alpha}{\alpha p^m(1-p)}. \quad \blacklozenge
 \end{aligned}$$

Example 10d A system has two components, the lives of which are independent exponential random variables with means $1/\lambda_1$ and $1/\lambda_2$, respectively. The system functions if and only if at least one of the components operates. For $i = 1, 2$, when component i fails, it is replaced with a new component that has the same lifetime distribution as the old one. The time it takes to replace component i is exponential with mean $1/\mu_i$ independently of the lifetimes of the components. If the system is presently functioning with both components operative, what is the expected value of the time until the system fails?

Solution: For $i = 1, 2$, let X_i be the lifetime of component i , and let T_i be the time it takes to replace component i when it fails. We say that the system is in *state 0* if none of its components operates, it is in *state 1* if its first component operates and its second component does not, it is in *state 2* if the second component operates but its first component does not, and, finally, it is in *state 3* if both of its components operate. For $i = 1, 2, 3$, let W_i be the time until the system fails given that it is currently in state i . We are interested in $E(W_3)$. Given that the system is in state 3, let N be the next state of the system. By Theorem 10.6 of the book,

$$\begin{aligned}
 E(W_3) &= E[E(W_3 | N)] \\
 &= E(W_3 | N = 1)P(N = 1) + E(W_3 | N = 2)P(N = 2). \quad (2)
 \end{aligned}$$

Considering the memoryless property of exponential random variables, by Remark 10.3 of the book,

$$\begin{aligned}
 E(W_3 | N = 1) &= E(X_2 | X_2 < X_1) + E(W_1) = \frac{1}{\lambda_1 + \lambda_2} + E(W_1) \\
 E(W_3 | N = 2) &= E(X_1 | X_1 < X_2) + E(W_2) = \frac{1}{\lambda_1 + \lambda_2} + E(W_2).
 \end{aligned}$$

Now, by Remark 10.2 of the book,

$$\begin{aligned}
 P(N = 1) &= P(X_2 < X_1) = \frac{\lambda_2}{\lambda_1 + \lambda_2}, \\
 P(N = 2) &= P(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}.
 \end{aligned}$$

Thus relation (2) gives

$$\begin{aligned} E(W_3) &= \left[\frac{1}{\lambda_1 + \lambda_2} + E(W_1) \right] \frac{\lambda_2}{\lambda_1 + \lambda_2} + \left[\frac{1}{\lambda_1 + \lambda_2} + E(W_2) \right] \frac{\lambda_1}{\lambda_1 + \lambda_2} \\ &= \frac{1}{\lambda_1 + \lambda_2} + \frac{\lambda_2}{\lambda_1 + \lambda_2} E(W_1) + \frac{\lambda_1}{\lambda_1 + \lambda_2} E(W_2). \end{aligned} \quad (3)$$

Now suppose that the system is in state 1. Letting N be the next state of the system, by Theorem 10.6 of the book,

$$\begin{aligned} E(W_1) &= E[E(W_1 | N)] \\ &= E(W_1 | N = 0)P(N = 0) + E(W_2 | N = 3)P(N = 3). \end{aligned} \quad (4)$$

Taking the memoryless property of exponential random variables into account, we have

$$\begin{aligned} E(W_1 | N = 0) &= E(X_1 | X_1 < T_2) = \frac{1}{\lambda_1 + \mu_2}, \\ E(W_1 | N = 3) &= E(T_2 | T_2 < X_1) + E(W_3) \\ &= \frac{1}{\lambda_1 + \mu_2} + E(W_3). \end{aligned}$$

Noting that

$$\begin{aligned} P(N = 0) &= P(X_1 < T_2) = \frac{\lambda_1}{\lambda_1 + \mu_2}, \\ P(N = 3) &= P(T_2 < X_1) = \frac{\mu_2}{\lambda_1 + \mu_2}, \end{aligned}$$

relation (4) gives

$$\begin{aligned} E(W_1) &= \frac{1}{\lambda_1 + \mu_2} \cdot \frac{\lambda_1}{\lambda_1 + \mu_2} + \left[\frac{1}{\lambda_1 + \mu_2} + E(W_3) \right] \cdot \frac{\mu_2}{\lambda_1 + \mu_2} \\ &= \frac{1}{\lambda_1 + \mu_2} + \frac{\mu_2}{\lambda_1 + \mu_2} E(W_3). \end{aligned} \quad (5)$$

By symmetry, we also have

$$E(W_2) = \frac{1}{\lambda_2 + \mu_1} + \frac{\mu_1}{\lambda_2 + \mu_1} E(W_3). \quad (6)$$

Substituting (5) and (6) in (3) and solving for $E(W_3)$ yields

$$E(W_3) = \frac{\lambda_1(\lambda_1 + \mu_2) + \lambda_2(\lambda_2 + \mu_1) + (\lambda_1 + \mu_2)(\lambda_2 + \mu_1)}{\lambda_1\lambda_2(\lambda_1 + \lambda_2 + \mu_1 + \mu_2)}. \quad \blacklozenge$$

Example 10e Let X be a binomial random variable with parameters n and p , where p itself is a random variable, which has a beta distribution with parameters α and β . Find the probability mass function of X .

Solution: Let f be the probability density function of p . By conditioning on p , for $0 \leq i \leq n$, we have

$$\begin{aligned}
 P(X = i) &= \int_0^1 P(X = i \mid p = x) f(x) dx \\
 &= \int_0^1 \left[\binom{n}{i} x^i (1-x)^{n-i} \right] \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1} dx \\
 &= \binom{n}{i} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x^{\alpha+i-1} (1-x)^{n-i+\beta-1} dx \\
 &= \binom{n}{i} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha + i)\Gamma(\beta + n - i)}{\Gamma(\alpha + \beta + n)}.
 \end{aligned}$$

Note that

$$\binom{n}{i} = \frac{\Gamma(n+1)}{\Gamma(i+1)\Gamma(n-i+1)}.$$

So

$$P(X = i) = \frac{\Gamma(n+1)\Gamma(\alpha + \beta)\Gamma(\alpha + i)\Gamma(\beta + n - i)}{\Gamma(i+1)\Gamma(n-i+1)\Gamma(\alpha)\Gamma(\beta)\Gamma(\alpha + \beta + n)}, \quad i = 0, 1, \dots, n. \quad \blacklozenge$$

Example 10f An Alternative Proof for Cauchy-Schwarz Inequality For random variables X and Y with finite variances,

$$|E(XY)| \leq \sqrt{E(X^2)E(Y^2)}.$$

Proof: For all real numbers t , $(X - tY)^2 \geq 0$. Hence, for all values of t ,

$$X^2 - 2XYt + t^2Y^2 \geq 0.$$

Since nonnegative random variables have nonnegative expected values,

$$E(X^2 - 2XYt + t^2Y^2) \geq 0,$$

which implies that

$$E(X^2) - 2E(XY)t + t^2E(Y^2) \geq 0.$$

Rewriting this as a polynomial in t of degree 2, we get

$$E(Y^2)t^2 - 2E(XY)t + E(X^2) \geq 0.$$

It is a well-known fact that if a polynomial of degree 2 is nonnegative, its discriminant is nonpositive. Therefore,

$$4[E(XY)]^2 - 4E(X^2)E(Y^2) \leq 0$$

or

$$[E(XY)]^2 \leq E(X^2)E(Y^2).$$

This gives

$$|E(XY)| \leq \sqrt{E(X^2)E(Y^2)}. \quad \blacklozenge$$

Example 10g A device has n components the lifetimes of which are independent exponential random variables with parameters $\lambda_1, \lambda_2, \dots, \lambda_n$, respectively. What is the probability that the i th component is the first to fail?

Solution: Let $X_1, X_2, \dots, X_n$ be the lifetimes of the components. Let

$$Y = \min(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n).$$

By Remark 9.2 of the book, Y is exponential with parameter $\lambda_1 + \dots + \lambda_{i-1} + \lambda_{i+1} + \dots + \lambda_n$.
By Remark 10.2 of the book, the desired probability is

$$P(\min(X_1, \dots, X_n) = X_i) = P(X_i < Y) = \frac{\lambda_i}{\lambda_1 + \lambda_2 + \dots + \lambda_n}.$$

Remark: Note that this example implies the following important property of the exponential random variables.

Let $X_1, X_2, \dots, X_n$ be $n > 1$ independent exponential random variables with parameters $\lambda_1, \lambda_2, \dots, \lambda_n$, respectively. Then

$$P(\min(X_1, X_2, \dots, X_n) = X_i) = \frac{\lambda_i}{\lambda_1 + \lambda_2 + \dots + \lambda_n}. \quad \blacklozenge$$

Example 10h Let $X_1, X_2, \dots, X_n$ be n independent exponential random variables with parameters $\lambda_1, \lambda_2, \dots, \lambda_n$, respectively. Let $X = \min(X_1, X_2, \dots, X_n)$, and define N to be i if and only if $X = X_i$. Therefore, N is a discrete random variable with the set of possible values $\{1, 2, \dots, n\}$ and probability mass function given by

$$P(N = i) = P(X = X_i) = \frac{\lambda_i}{\lambda_1 + \lambda_2 + \dots + \lambda_n}.$$

(See Example 10g.) Show that X and N are independent random variables.

Solution: It suffices to show that

$$P(X > t, N = i) = P(X > t)P(N = i).$$

We have

$$\begin{aligned}
 P(X > t, N = i) &= P(X > t, X = X_i) \\
 &= \int_0^\infty P(X > t, X = X_i \mid X_i = x) \lambda_i e^{-\lambda_i x} dx \\
 &= \int_t^\infty P(X = X_i \mid X_i = x) \lambda_i e^{-\lambda_i x} dx \\
 &= \int_t^\infty P(X_1 \geq x, \dots, X_{i-1} \geq x, X_{i+1} \geq x, \dots, X_n \geq x \mid X_i = x) \lambda_i e^{-\lambda_i x} dx \\
 &= \int_t^\infty P(X_1 \geq x) \cdots P(X_{i-1} \geq x) P(X_{i+1} > x) \cdots P(X_n \geq x) \lambda_i e^{-\lambda_i x} dx \\
 &= \lambda_i \int_t^\infty e^{-\lambda_1 x} \cdots e^{-\lambda_{i-1} x} \cdots e^{-\lambda_{i+1} x} \cdots e^{-\lambda_n x} \cdot e^{-\lambda_i x} dx \\
 &= \frac{\lambda_i}{\lambda_1 + \lambda_2 + \cdots + \lambda_n} \cdot e^{-(\lambda_1 + \lambda_2 + \cdots + \lambda_n)t} \\
 &= P(X = X_i) P(X > t) = P(N = i) P(X > t). \quad \blacklozenge
 \end{aligned}$$

Example 10i Suppose that there are k , $k > 1$, types of defects, identified by a wire manufacturing company, that occur independently over the cable as it comes out of a machine. For $1 \leq i \leq k$, the number of defects of type i is distributed according to a Poisson process with parameter λ_i . Find the probability that the n th defect occurring over the cable is of type i , $1 \leq i \leq k$.

Solution: For $1 \leq j \leq k$, let V_j be the number of meters on the cable until the first defect of type j occurs. Note that $P(V_i = \min(V_1, V_2, \dots, V_k))$ is the probability that the first defect occurring over the cable is of type i . Suppose that the first defect occurs when exactly u meters of the cable are produced. That point labeled $t = 0$, from there on, by stationarity and independent-increment property, probabilistically, the behaviors of the k given Poisson processes are identical to the original processes when the first u meters of cable wire were being produced. So the probability that the second defect is of type i is identical to the probability that the first defect is of type i and so on. Hence, by Example 10g, the desired quantity is

$$P(V_i = \min(V_1, V_2, \dots, V_k)) = \frac{\lambda_i}{\lambda_1 + \lambda_2 + \cdots + \lambda_k},$$

since V_i 's are independent exponential random variables with parameters $\lambda_1, \lambda_2, \dots, \lambda_k$, respectively.

Example 10j (Investment) Dr. Caprio has invested money in three uncorrelated assets; 25% in the first one, 43% in the second one, and 32% in the third one. The means of the annual rate of returns for these assets, respectively, are 10%, 15%, and 13%. Their standard deviations are 8%, 12%, and 10%, respectively. Find the mean and standard deviation of the annual rate of return for Dr. Caprio's total investment.

Solution: Let r be the annual rate of return for Dr. Caprio's total investment. Let r_1 , r_2 , and r_3 be the annual rate of returns for the first, second, and third assets, respectively. By relation (4) of Section 4B,

$$r = 0.25r_1 + 0.43r_2 + 0.32r_3.$$

Thus

$$\begin{aligned} E(r) &= 0.25E(r_1) + 0.43E(r_2) + 0.32E(r_3) \\ &= (0.25)(0.10) + (0.43)(0.15) + (0.32)(0.13) = 0.1311. \end{aligned}$$

Since the assets are uncorrelated, by relation (10.11) of the book,

$$\begin{aligned} \text{Var}(r) &= (0.25)^2 \text{Var}(r_1) + (0.43)^2 \text{Var}(r_2) + (0.32)^2 \text{Var}(r_3) \\ &= (0.25)^2(0.08)^2 + (0.43)^2(0.12)^2 + (0.32)^2(0.10)^2 = 0.004087. \end{aligned}$$

Therefore, $\sigma_r = \sqrt{0.004087} = 0.064$. Hence Dr. Caprio should expect an annual rate of return of 13.11% with standard deviation 6.4%. Note that Dr. Caprio has reduced the standard deviation of his investments considerably by diversifying his investment; that is, by not putting all of his eggs in one basket.

In general,

If Dr. Caprio divides his money into n equal portions and invests them in n uncorrelated financial assets with identically distributed rates of returns $r_1, r_2, \dots, r_n$, each with variance σ^2 , then by (10.11),

$$\begin{aligned} \text{Var}(r) &= \text{Var}\left(\frac{1}{n}r_1 + \frac{1}{n}r_2 + \dots + \frac{1}{n}r_n\right) \\ &= \frac{1}{n^2}\text{Var}(r_1) + \frac{1}{n^2}\text{Var}(r_2) + \dots + \frac{1}{n^2}\text{Var}(r_n) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}. \end{aligned}$$

This shows that the more financial assets Dr. Caprio invests in, the less the variance of the return will be—an indication that diversification reduces the investment risks. ♦

Example 10k (Investment) Dr. Siciliano has invested money in two financial assets. Let r be the annual rate of return for his total investment. Let r_1 and r_2 be the annual rates of return for the first and second assets, respectively. Let $\sigma^2 = \text{Var}(r)$, $\sigma_1^2 = \text{Var}(r_1)$, and $\sigma_2^2 = \text{Var}(r_2)$. Prove that

$$\sigma^2 \leq \max(\sigma_1^2, \sigma_2^2).$$

In particular, by this inequality, if r_1 and r_2 are identically distributed, then $\sigma_1^2 = \sigma_2^2$ implies that

$$\sigma^2 \leq \sigma_1^2 = \sigma_2^2.$$

This is an extension of the result discussed in Example 10j. It shows that, even if the financial assets are correlated, still diversification reduces the investment risk.

Proof: We will show that if $\sigma_2^2 \leq \sigma_1^2$, then $\sigma^2 \leq \sigma_1^2$. By symmetry, if $\sigma_1^2 \leq \sigma_2^2$, then $\sigma^2 \leq \sigma_2^2$. Therefore, σ^2 is either less than or equal to σ_1^2 , or it is less than or equal to σ_2^2 , implying that

$$\sigma^2 \leq \max(\sigma_1^2, \sigma_2^2).$$

To show that $\sigma_2^2 \leq \sigma_1^2$ implies that $\sigma^2 \leq \sigma_1^2$, let w_1 and w_2 be the fractions of Dr. Siciliano's investment in the first and second financial assets, respectively. Then $w_1 + w_2 = 1$ and

$$r = w_1 r_1 + w_2 r_2.$$

Let ρ be the correlation coefficient of r_1 and r_2 . Then

$$\text{Cov}(r_1, r_2) = \rho \sigma_1 \sigma_2.$$

We have

$$\begin{aligned} \text{Var}(r) &= \text{Var}(w_1 r_1 + w_2 r_2) = \text{Var}(w_1 r_1) + \text{Var}(w_2 r_2) + 2\text{Cov}(w_1 r_1, w_2 r_2) \\ &= w_1^2 \text{Var}(r_1) + w_2^2 \text{Var}(r_2) + 2w_1 w_2 \text{Cov}(r_1, r_2). \end{aligned}$$

Noting that $w_1 + w_2 = 1$, $-1 \leq \rho \leq 1$, and $\sigma_2^2 \leq \sigma_1^2$, this relation implies that

$$\begin{aligned} \sigma^2 &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2 \\ &\leq w_1^2 \sigma_1^2 + w_2^2 \sigma_1^2 + 2w_1 w_2 \sigma_1^2 \\ &= (w_1^2 + w_2^2 + 2w_1 w_2) \sigma_1^2 \\ &= (w_1 + w_2)^2 \sigma_1^2 = \sigma_1^2. \quad \blacklozenge \end{aligned}$$

Example 10ℓ (Investment) Mr. Ingham has invested money in three assets; 18% in the first asset, 40% in the second one, and 42% in the third one. Let r_1 , r_2 , and r_3 be the annual rate of returns for these three investments, respectively. For $1 \leq i, j \leq 3$, $\text{Cov}(r_i, r_j)$ is the i th element in the j th row of the following table. [Note that $\text{Var}(r_i) = \text{Cov}(r_i, r_i)$.]

	r_1	r_2	r_3
r_1	0.064	0.03	0.015
r_2	0.03	0.0144	0.021
r_3	0.015	0.021	0.01

Find the standard deviation of the annual rate of return for Mr. Ingham's total investment.

Solution: Let r be the annual rate of return for Mr. Ingham's total investment. We have

$$\begin{aligned} \text{Var}(r) &= \text{Var}(0.18r_1 + 0.40r_2 + 0.42r_3) \\ &= (0.18)^2 \text{Var}(r_1) + (0.40)^2 \text{Var}(r_2) + (0.42)^2 \text{Var}(r_3) \\ &\quad + 2(0.18)(0.40)\text{Cov}(r_1, r_2) + 2(0.18)(0.42)\text{Cov}(r_1, r_3) \\ &\quad + 2(0.40)(0.42)\text{Cov}(r_2, r_3) \\ &= (0.18)^2(0.064) + (0.40)^2(0.0144) + (0.42)^2(0.01) \\ &\quad + 2(0.18)(0.40)(0.03) + 2(0.18)(0.42)(0.015) + 2(0.40)(0.42)(0.021) \\ &= 0.01979. \end{aligned}$$

Hence the standard deviation of the annual rate of return for Mr. Ingham's total investment is $\sqrt{0.01979} = 0.14$. ♦

Example 10m (Investment) Mr. Kowalski has invested \$50,000 in three uncorrelated financial assets: 25% in the first financial asset, 40% in the second one, and 35% in the third one. The annual rates of return for these assets, respectively, are normal random variables with means 12%, 15%, and 18%, and standard deviations 8%, 12%, and 15%. What is the probability that, after a year, Mr. Kowalski's net profit from this investment is at least \$10,000? Ignore transaction costs and assume that there is no annual dividend.

Solution: Let r_1 , r_2 , and r_3 be the annual rates of return for Mr. Kowalski's investments in financial assets 1, 2, and 3, respectively. Let r be the annual rate of return for his total investment. Then, by relation (4) of Section 4B,

$$r = 0.25r_1 + 0.40r_2 + 0.35r_3.$$

Since the assets are uncorrelated, we have

$$E(r) = (0.25)(0.12) + (0.40)(0.15) + (0.35)(0.18) = 0.153,$$

$$\text{Var}(r) = (0.25)^2(0.08)^2 + (0.40)^2(0.12)^2 + (0.35)^2(0.15)^2 = 0.00546,$$

$$\sigma_r = \sqrt{\text{Var}(r)} = 0.074.$$

Hence $r \sim N(0.153, 0.00546)$. Let X be the total investment of Mr. Kowalski. We are given that $X = 50,000$. Let Y be the total return of Mr. Kowalski's investment next year. The desired probability is

$$\begin{aligned} P(Y - X \geq 10,000) &= P\left(\frac{Y - X}{X} \geq \frac{10,000}{50,000}\right) \\ &= P(r \geq 0.2) = P\left(Z \geq \frac{0.2 - 0.153}{0.074}\right) \\ &= P(Z \geq 0.64) = 1 - \Phi(0.64) = 1 - 0.7389 = 0.2611. \quad \blacklozenge \end{aligned}$$

Example 10n (Genetics) Hemophilia is a sex-linked disease with normal allele H dominant to the mutant allele h . Kim and John are married, and John is phenotypically normal. Suppose that, in the entire population, the frequencies of H and h are 0.98 and 0.02, respectively. If Kim and John have four sons and three daughters, what is the expected number of their hemophilic children?

Solution: Since hemophilia is a sex-linked disease, and John is phenotypically normal, John is H . Therefore, no matter what Kim's genotype is, none of the daughters has hemophilia. Whether a boy has hemophilia or not depends solely on the genotype of Kim. Let X be the number of the boys who have hemophilia. To find, $E(X)$, the expected number of the boys who have hemophilia, let

$$Z = \begin{cases} 0 & \text{if Kim is } hh \\ 1 & \text{if Kim is } Hh \\ 2 & \text{if Kim is } HH. \end{cases}$$

Then

$$\begin{aligned} E(X) &= E[E(X \mid Z)] \\ &= E(X \mid Z = 0)P(Z = 0) + E(X \mid Z = 1)P(Z = 1) + E(X \mid Z = 2)P(Z = 2) \\ &= 4(0.02)(0.02) + 4(1/2)[2(0.98)(0.02)] + 0[0.98)(0.98)] = 0.08. \end{aligned}$$

Therefore, on average, 0.08 of the boys and hence 0.08 of the children are expected to have hemophilia. ♦

10B PATTERN APPEARANCE

Suppose that a coin is tossed independently and successively. We are interested in the expected number of tosses until a specific pattern is first obtained. For example, the expected number of tosses until the first appearance of HT, the first appearance of HHH, or the first appearance of, say, HHTHH. Similarly, suppose that in generating random numbers from the set $\{0, 1, 2, \dots, 9\}$ independently and successively, we are interested in the expected number of digits to be generated until the first appearance of 453, the first appearance of 353, or the first appearance of, say, 88588.

In coin tossing, for example, we can readily calculate the expected number of tosses after the first appearance of a pattern until its second appearance. The same is true in generating random numbers and other similar experiments. As an example, shortly, we will show that it is not difficult to calculate the expected number of digits to be generated after the first appearance of a pattern until its second appearance. We use this to calculate the expected number of digits to be generated until the first appearance of the pattern. First, we introduce some notation and make some definitions:

Suppose that A is a pattern of certain characters. Let the number of random characters generated until A appears for the first time be denoted by $\rightarrow A$. Let the number of random characters generated after the appearance of a pattern A until the next appearance of a pattern B be denoted by $A \rightarrow B$.

Definition 10a Suppose that A is a pattern of length n of certain characters. For $1 \leq i < n$, let $A_{(i)}$ and $A^{(i)}$ be the first i and the last i characters of A , respectively. If $A_{(i)} = A^{(i)}$ for some i ($1 \leq i < n$), we say that the pattern is **overlapping**. Otherwise, we say it is a pattern with **no self-overlap**. For an overlapping pattern A , if $A_{(i)} = A^{(i)}$, then i is said to be an **overlap number**.

For example, in successive and independent tosses of a coin, let $A = \text{HHTHH}$; we have

$$A_{(1)} = \text{H}, \quad A_{(2)} = \text{HH}, \quad A_{(3)} = \text{HHT}, \quad A_{(4)} = \text{HHTH};$$

$$A^{(1)} = \text{H}, \quad A^{(2)} = \text{HH}, \quad A^{(3)} = \text{THH}, \quad A^{(4)} = \text{HTHH}.$$

We see that A is overlapping with overlap numbers 1 and 2. As another example, in generating random digits from the set $\{0, 1, 2, \dots, 9\}$ independently and successively, consider the pattern $A = 45345$; we have

$$A_{(1)} = 4, \quad A_{(2)} = 45, \quad A_{(3)} = 453, \quad A_{(4)} = 4534;$$

$$A^{(1)} = 5, \quad A^{(2)} = 45, \quad A^{(3)} = 345, \quad A^{(4)} = 5345.$$

Since $A_{(2)} = A^{(2)}$, the pattern A is overlapping, and 2 is its only overlap number. As a third example, the pattern 33333 is overlapping. Its overlap numbers are 1, 2, 3, 4, and 5. The patterns 3334 and 44565 are patterns with no self-overlap.

Next, we make the following observations: For a pattern with no self-overlap such as 453, $E(\rightarrow 453) = E(453 \rightarrow 453)$. That is, the expected number of random numbers to be generated until the first 453 appears is the same as the expected number of random numbers to be generated after the first appearance of 453 until its second appearance. However, this does not apply to an overlapping pattern such as 353. To our surprise, $E(\rightarrow 353)$ is *larger* than $E(353 \rightarrow 353)$. This is because $E(353 \rightarrow 353) = E(3 \rightarrow 353)$ and

$$\begin{aligned} E(\rightarrow 353) &= E(\rightarrow 3) + E(3 \rightarrow 353) \\ &= E(\rightarrow 3) + E(353 \rightarrow 353). \end{aligned}$$

That is, the expected number of digits to be generated until the pattern 353 appears is the expected number of digits to be generated until the first appearance of 3, plus the expected number of additional digits after that until 353 is obtained for the first time. In general, *for patterns of the same length, on average, overlapping ones occur later than those with no self-overlap*. For example, when random digits are generated one after another independently and successively, on average, 453 appears sooner than 353, and 353 appears sooner than 333. Note that

$$\begin{aligned} E(\rightarrow 333) &= E(\rightarrow 3) + E(3 \rightarrow 33) + E(33 \rightarrow 333) \\ &= E(\rightarrow 3) + E(33 \rightarrow 33) + E(333 \rightarrow 333). \end{aligned}$$

In general, let A be an overlapping pattern. Let $i_1, i_2, i_3, \dots, i_m$ be the overlap numbers of A arranged in (strictly) ascending order. Then

$$\begin{aligned} E(\rightarrow A) &= E(\rightarrow A_{(i_1)}) + E(A_{(i_1)} \rightarrow A_{(i_2)}) + E(A_{(i_2)} \rightarrow A_{(i_3)}) \\ &\quad + \dots + E(A_{(i_m)} \rightarrow A). \end{aligned}$$

Clearly, the number of digits to be generated to obtain, say, the first 3, the first 5, or the first 8 are all geometric random variables with parameter $p = 1/10$. So

$$E(\rightarrow 3) = E(\rightarrow 5) = E(\rightarrow 8) = \frac{1}{p} = 10.$$

In general, for a pattern A , this and $E(A \rightarrow A)$ enable us to find $E(\rightarrow A)$. So all that remains is to find the expected number of digits to be generated after the first appearance of a pattern until its second appearance. To demonstrate one method of calculation, we will find $E(353 \rightarrow 353)$ using an *intuitive probabilistic* argument. To do so, in the process of generating random digits, let D_1 be the first random digit and D_2 be the first two random digits generated. For $i \geq 3$, let D_i be the $(i-2)$ nd, $(i-1)$ st, and i th random digits generated. For example, if the first 6 random digits generated are 507229, then $D_1 = 5$, $D_2 = 50$, $D_3 = 507$, $D_4 = 072$, $D_5 = 722$, and $D_6 = 229$. Let $X_1 = X_2 = 0$. For $i \geq 3$, let

$$X_i = \begin{cases} 1 & \text{if } D_i = 353 \\ 0 & \text{if } D_i \neq 353. \end{cases}$$

Then the number of appearances of 353 among the first n random digits generated is $\sum_{i=1}^n X_i$. Hence the proportion of the D_i 's that are 353 is $(1/n) \sum_{i=1}^n X_i$ when n random digits are generated. Now $E(X_1) = E(X_2) = 0$, and for $i \geq 3$, $E(X_i) = E(X_3)$

since the X_i 's are identically distributed. Thus the expected value of the fraction of D_i 's that are 353 is

$$\begin{aligned} E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) &= \frac{1}{n} E\left(\sum_{i=1}^n X_i\right) = \frac{1}{n} \left[\sum_{i=1}^n E(X_i) \right] = \frac{1}{n} \cdot (n-2) E(X_3) \\ &= \frac{n-2}{n} P(D_3 = 353) = \frac{n-2}{n} \left(\frac{1}{10}\right)^3 = \frac{n-2}{1000n}. \end{aligned}$$

Thus the expected value of the fraction of D_i 's in n random digits that are 353 is $(n-2)/(1000n)$. Now as $n \rightarrow \infty$, this expected value approaches $1/1000$, implying that in the long-run, on average, the fraction of D_i 's that are 353 is $1/1000$. This and the fact that the expected number of random digits between any two consecutive 353's is the same, imply that the average number of random digits between two consecutive 353's is 1000. Hence $E(353 \rightarrow 353) = 1000$. A rigorous proof for this fact needs certain theorems of *renewal theory*, a branch of stochastic processes. See, for example, *Stochastic Modeling and the Theory of Queues*, by Ronald W. Wolff, Prentice Hall, 1989. We can use the argument above to show that, for example,

$$\begin{aligned} E(23 \rightarrow 23) &= 100, \\ E(453 \rightarrow 453) &= 1000, \\ E(333 \rightarrow 333) &= 1000, \\ E(5732 \rightarrow 5732) &= 10,000. \end{aligned}$$

Therefore,

$$E(\rightarrow 453) = E(453 \rightarrow 453) = 1000,$$

$$\begin{aligned} E(\rightarrow 353) &= E(\rightarrow 3) + E(3 \rightarrow 353) \\ &= E(\rightarrow 3) + E(353 \rightarrow 353) \\ &= 10 + 1000 = 1010, \end{aligned}$$

$$\begin{aligned} E(\rightarrow 333) &= E(\rightarrow 3) + E(3 \rightarrow 33) + E(33 \rightarrow 333) \\ &= E(\rightarrow 3) + E(33 \rightarrow 33) + E(333 \rightarrow 333) \\ &= 10 + 100 + 1000 = 1110, \end{aligned}$$

and

$$\begin{aligned} E(\rightarrow 88588) &= E(\rightarrow 8) + E(8 \rightarrow 88) + E(88 \rightarrow 88588) \\ &= E(\rightarrow 8) + E(88 \rightarrow 88) + E(88588 \rightarrow 88588) \\ &= 10 + 100 + 100000 = 100110. \end{aligned}$$

An argument similar to the preceding will establish the following generalization.

Suppose that an experiment results in one of the outcomes $a_1, a_2, \dots, a_k$, and $P(\{a_i\}) = p_i$, $i = 1, 2, \dots, k$; $\sum_{i=1}^k p_i = 1$. Let $a_{i_1}, a_{i_2}, \dots, a_{i_\ell}$, be (not necessarily distinct) elements of $\{a_1, a_2, \dots, a_k\}$. Then in successive and independent performances of this experiment, the expected number of trials after the first appearance of the pattern $a_{i_1}a_{i_2}\cdots a_{i_\ell}$ until its second appearances is $1/(p_{i_1}p_{i_2}\cdots p_{i_\ell})$.

By this generalization, for example, in successive independent flips of a fair coin,

$$\begin{aligned} E(\rightarrow HH) &= E(\rightarrow H) + E(H \rightarrow HH) \\ &= E(\rightarrow H) + E(HH \rightarrow HH) \\ &= \frac{1}{1/2} + \frac{1}{(1/2)(1/2)} = 6, \end{aligned}$$

whereas

$$E(\rightarrow HT) = E(HT \rightarrow HT) = \frac{1}{(1/2)(1/2)} = 4.$$

Similarly,

$$\begin{aligned} E(\rightarrow HHHH) &= E(\rightarrow H) + E(H \rightarrow HH) + E(HH \rightarrow HHH) \\ &\quad + E(HHH \rightarrow HHHH) \\ &= E(\rightarrow H) + E(HH \rightarrow HH) + E(HHH \rightarrow HHH) \\ &\quad + E(HHHH \rightarrow HHHH) \\ &= 2 + 4 + 8 + 16 = 30, \end{aligned}$$

whereas

$$E(\rightarrow THHH) = E(THHH \rightarrow THHH) = 16.$$

At first glance, it seems paradoxical that, on the average, it takes nearly twice as many flips of a fair coin to obtain the first HHHH as to encounter THHH for the first time. However, THHH is a pattern with no self-overlap, whereas HHHH is an overlapping pattern.

EXERCISES FOR SECTION 10B

- Suppose that random digits are generated from the set $\{0, 1, \dots, 9\}$ independently and successively. Find the expected number of digits to be generated until the pattern (a) 007 appears, (b) 156156 appears, (c) 575757 appears.
- In successive independent flips of a fair coin, what is the expected number of trials until the pattern THTHTTHTHT appears?



10C MORE ON MEMORYLESS PROPERTY OF EXPONENTIAL RANDOM VARIABLES

In this section, we will discuss an aspect of the exponential random variables, which demonstrates the validity of memoryless property even in more complicated situations. Suppose that a system has two devices, the lifetime of which, X_1 and X_2 , are independent exponential random variables with means $1/\lambda_1$ and $1/\lambda_2$, respectively. We will show that, given device 1 fails before device 2, the remaining lifetime of device 2 is still exponential with the mean $1/\lambda_2$ independently of the lifetime of device 1. This property has many useful applications and is the subject of the following theorem.

Theorem 10a *Let X_1 and X_2 be independent exponential random variables with parameters λ_1 and λ_2 , respectively. Then, for $t \geq 0$,*

$$P(X_2 - X_1 > t \mid X_2 > X_1) = e^{-\lambda_2 t};$$

That is, given $X_2 > X_1$, the random variable $X_2 - X_1$ is exponential with parameter λ_2 . Furthermore, $X_2 - X_1$ and X_1 are independent random variables.

Proof: Using the definition of conditional probability and Remark 10.2 of the book yields

$$\begin{aligned} P(X_2 - X_1 > t \mid X_2 > X_1) &= \frac{P(X_2 - X_1 > t, X_2 > X_1)}{P(X_2 > X_1)} = \frac{P(X_2 > X_1 + t)}{\lambda_1 / (\lambda_1 + \lambda_2)} \\ &= \frac{\lambda_1 + \lambda_2}{\lambda_1} \int_0^\infty P(X_2 > X_1 + t \mid X_1 = x) \lambda_1 e^{-\lambda_1 x} dx \\ &= \frac{\lambda_1 + \lambda_2}{\lambda_1} \int_0^\infty P(X_2 > x + t) \lambda_1 e^{-\lambda_1 x} dx \\ &= \frac{\lambda_1 + \lambda_2}{\lambda_1} \int_0^\infty e^{-\lambda_2(x+t)} \cdot \lambda_1 e^{-\lambda_1 x} dx \\ &= \frac{\lambda_1 + \lambda_2}{\lambda_1} \cdot \frac{\lambda_1}{\lambda_1 + \lambda_2} e^{-\lambda_2 t} = e^{-\lambda_2 t}. \end{aligned}$$

The following shows that given $X_2 > X_1$, the random variables $X_2 - X_1$ and X_1 are independent.

$$\begin{aligned} P(X_2 - X_1 > t_2 \mid X_2 > X_1, X_1 = t_1) &= \frac{P(X_2 - X_1 > t_2, X_2 > X_1 \mid X_1 = t_1)}{P(X_2 > X_1 \mid X_1 = t_1)} \\ &= \frac{P(X_2 > t_1 + t_2)}{P(X_2 > t_1)} = \frac{e^{-\lambda_2(t_1+t_2)}}{e^{-\lambda_2 t_1}} = e^{-\lambda_2 t_2} \\ &= P(X_2 - X_1 > t_2 \mid X_2 > X_1). \quad \blacklozenge \end{aligned}$$

Theorem 10a can be generalized in the following way. Suppose that a system has n ($n > 2$) devices, the lifetime of which, $X_1, X_2, \dots, X_n$, are independent exponential random variables with means $1/\lambda_1, 1/\lambda_2, \dots, 1/\lambda_n$, respectively. Suppose that we are given that device 1 is the first device to fail. Given this information, $X_2 - X_1, X_3 - X_1$,

$\dots, X_n - X_1$ are $n - 1$ independent exponential random variables with means $1/\lambda_2, 1/\lambda_3, \dots, 1/\lambda_n$, respectively. Furthermore, $X_i - X_1$ is independent of X_1 for $2 \leq i \leq n$.

Example 10o Upon arrival at a shoe store, the times it will take Sara and Martha to buy a pair of shoes are independent exponential random variables with means $1/\lambda$ and $1/\mu$, respectively. If Sara and Martha enter the shoe store at the same time, find the expected length of time until they have both purchased pairs of shoes if (a) $1/\lambda = 1/\mu = 30$ minutes; (b) $1/\lambda = 30$ minutes, but $1/\mu = 45$ minutes.

Solution: Let X be the time it will take for Sara to buy a pair of shoes, and let Y be the time it will take for Martha to buy a pair of shoes. Let S_1 be the time when exactly one of Sara or Martha has purchased a pair of shoes, and S_2 be the time when both of them have purchased shoes.

- (a) By Remark 9.2 of the book, $E(S_1) = E[\min(X, Y)] = \frac{1}{(1/30) + (1/30)} = 15$ minutes. At S_1 , by Theorem 10a, for whoever is not done buying shoes yet, the remaining time to complete her purchase is exponential with mean 30 minutes. Hence $E(S_2 - S_1) = 30$ minutes and thus

$$E(S_2) = E(S_1) + E(S_2 - S_1) = 45 \text{ minutes.}$$

- (b) In this case, $E(S_1) = E[\min(X, Y)] = \frac{1}{(1/30) + (1/45)} = 18$ minutes. To find $E(S_2 - S_1)$, by Theorem 10.10 of the book,

$$\begin{aligned} E(S_2 - S_1) &= E(S_2 - S_1 \mid X < Y)P(X < Y) + E(S_2 - S_1 \mid Y < X)P(Y < X) \\ &= E(Y - X \mid X < Y)P(X < Y) + E(X - Y \mid Y < X)P(Y < X) \\ &= 45 \cdot \frac{1/30}{(1/30) + (1/45)} + 30 \cdot \frac{(1/45)}{(1/30) + (1/45)} = 39 \text{ minutes.} \end{aligned}$$

Hence

$$E(S_2) = E(S_1) + E(S_2 - S_1) = 18 + 39 = 57 \text{ minutes.}$$

Remark: An alternative solution to this problem is to note that $S_2 = \max(X, Y)$. Hence, by Remark 6.4 of the book,

$$\begin{aligned} E(S_2) &= E[\max(X, Y)] = \int_0^\infty P(\max(X, Y) > t) dt \\ &= \int_0^\infty [1 - P(\max(X, Y) \leq t)] dt = \int_0^\infty [1 - P(X \leq t, Y \leq t)] dt \\ &= \int_0^\infty [1 - P(X \leq t)P(Y \leq t)] dt = \int_0^\infty [(1 - e^{-t/30})(1 - e^{-t/45})] dt \\ &= 57 \text{ minutes. } \blacklozenge \end{aligned}$$

Example 10p To ship merchandise abroad, by freight, customers must go through two stations sequentially. At the first station, it takes a customer an exponential time with mean $1/\alpha$ to get clearance from the customs. At the second station, it takes a customer an exponential time with mean $1/\beta$ to repackage the merchandise and deliver it to the carrier for shipment. The service times at stations 1 and 2 are independent sequences of independent random variables. Suppose that when Mary arrives to ship her antique chair, Alan is being served in the customs, but there are no customers in the queue or in the second station. Find Mary's expected waiting time.

Solution: Let X_1 and X_2 be Mary's service times at stations 1 and 2, respectively. Let R be the remaining service time of Alan at station 1 when Mary arrives, and Y be Alan's service time at station 2. Due to memoryless property of exponential random variables, R is exponential with mean $1/\alpha$. Let V be the period between completion times of Mary's service at stations 1 and 2. Let W be Mary's total waiting time. Clearly,

$$\begin{aligned} E(W) &= E(R + X_1 + V) = E(R) + E(X_1) + E(V) \\ &= \frac{1}{\alpha} + \frac{1}{\alpha} + E(V) = \frac{2}{\alpha} + E(V) \end{aligned}$$

To find $E(V)$, note that, by Theorem 10.10 of the book,

$$\begin{aligned} E(V) &= E(V | Y < X_1)P(Y < X_1) + E(V | Y > X_1)P(Y > X_1) \\ &= E(X_2)P(Y < X_1) + [E(Y - X_1 | Y > X_1) + E(X_2)]P(Y > X_1). \end{aligned}$$

By Theorem 10a, given $Y > X_1$, $Y - X_1$ is exponential with mean $1/\beta$. Hence

$$E(V) = \frac{1}{\beta} \cdot \frac{\beta}{\alpha + \beta} + \left(\frac{1}{\beta} + \frac{1}{\beta} \right) \frac{\alpha}{\alpha + \beta} = \frac{\beta + 2\alpha}{\beta(\alpha + \beta)}.$$

The final answer is

$$E(W) = \frac{2}{\alpha} + E(V) = \frac{2}{\alpha} + \frac{\beta + 2\alpha}{\beta(\alpha + \beta)}. \quad \blacklozenge$$

Example 10q After a long day of work, the tellers of a seven-teller bank are anxious to leave for the day. However, at 5:00 P.M. when the bank closes, there are seven customers still in the bank being served. Luckily, there is no customer in the queue waiting to be served. If the service times are identically distributed, independent exponential random variables, each with mean 10 minutes, at what time should we expect the last teller to leave the bank? Assume that a teller who has served his or her last customer leaves for the day.

Solution: For the tellers' departure times, suppose that the i th departure occurs Y_i minutes past 5:00 P.M. Clearly, Y_1 is the minimum of seven independent exponential random variables each with parameter $\lambda = 1/10$. Thus, by Remark 9.2 of the book, Y_1 is exponential with parameter $7/10$, and hence $E(Y_1) = 10/7$ minutes. By Theorem 10a, at Y_1 minutes past 5:00 P.M., the remaining service times of the remaining six customers are independent

exponential random variables each with parameter $1/10$. Therefore, $Y_2 - Y_1$ is the minimum of six independent exponential random variables each with parameter $1/10$. Hence $Y_2 - Y_1$ is exponential with parameter $6/10$, and hence $E(Y_2 - Y_1) = 10/6$ minutes. Applying Theorem 10a again, at Y_2 minutes past 5:00 P.M., the remaining service times of the remaining five customers are independent exponential random variables each with parameter $1/10$. Thus $Y_3 - Y_2$ is the minimum of five independent exponential random variables each with parameter $1/10$, implying that $Y_3 - Y_2$ is exponential with parameter $5/10$. Hence $E(Y_3 - Y_2) = 10/5$ minutes. Continuing this argument, we have that, for $1 \leq i \leq 6$, $E(Y_{i+1} - Y_i) = 10/(7 - i)$. Thus

$$\begin{aligned} E(Y_7) &= E(Y_1) + E(Y_2 - Y_1) + \cdots + E(Y_6 - Y_5) + E(Y_7 - Y_6) \\ &= \frac{10}{7} + \frac{10}{6} + \cdots + \frac{10}{2} + 10 = \frac{363}{14}, \end{aligned}$$

which shows that the expected departure time for the last teller for the day is approximately 25 minutes and 56 seconds past 5:00 P.M. ♦

Example 10r The Dean of a college buys an 8-sheet crosscut paper shredder for his own office, a 12-sheet crosscut paper shredder for the Center for Writing and Reading (CWR), and a 15-sheet crosscut paper shredder for the main office. Suppose that, under normal usage, which is the case at this College, the lifetimes of these shredders are independent exponential random variables with means 6, 8, and 10 years, respectively. What is the probability that the Dean's own shredder dies first, the CWR's shredder dies next, and the main office's shredder dies last?

Solution: Let X_1 be the lifetime of the Dean's own shredder, X_2 be the lifetime of the CWR's shredder, and X_3 be the lifetime of the main office's shredder. We are given that X_1 , X_2 and X_3 are independent exponential random variables with parameters $\lambda_1 = 1/6$, $\lambda_2 = 1/8$, and $\lambda_3 = 1/10$, respectively. To calculate the desired probability

$$P(X_1 < X_2 < X_3),$$

note that, by Example 10g and Theorem 10a,

$$\begin{aligned} P(X_1 < X_2 < X_3) &= \\ P(X_2 < X_3 \mid \min(X_1, X_2, X_3) = X_1) &P(\min(X_1, X_2, X_3) = X_1) = \\ P(X_2 < X_3 \mid X_1 < X_2, X_1 < X_3) &\frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3} = \\ P(X_2 - X_1 < X_3 - X_1 \mid X_2 - X_1 > 0, &X_3 - X_1 > 0) \frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3}. \end{aligned}$$

We know that given $X_2 - X_1 > 0$ and $X_3 - X_1 > 0$, $X_2 - X_1$ and $X_3 - X_1$ are independent exponential random variables with parameters λ_2 and λ_3 , respectively. Hence

$$P(X_2 - X_1 < X_3 - X_1 \mid X_2 - X_1 > 0, X_3 - X_1 > 0) = \frac{\lambda_2}{\lambda_2 + \lambda_3}.$$

Therefore,

$$\begin{aligned} P(X_1 < X_2 < X_3) &= \frac{\lambda_2}{\lambda_2 + \lambda_3} \cdot \frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3} \\ &= \frac{1/8}{(1/8) + (1/10)} \cdot \frac{1/6}{(1/6) + (1/8) + (1/10)} = \frac{100}{423} = 0.236. \quad \blacklozenge \end{aligned}$$

10D CONDITIONING ON MORE THAN ONE RANDOM VARIABLE

We will now generalize the definition of $E(X|Y)$ to the multivariate case.

Definition 10b Let X , Y , and Z be three random variables. By the random variable $E(X | Y, Z)$ we mean a function of the random variables Y and Z that is defined to be $E(X | Y = y, Z = z)$ when $Y = y$ and $Z = z$.

Let p be the joint probability mass function of three discrete random variables X , Y and Z . Let $p_{X|Y,Z}(x|y, z)$ be the conditional probability mass function of X given that $Y = y$ and $Z = z$. Let $p_{Y,Z}$ be the joint probability mass function marginalized over Y and Z ; that is,

$$p_{Y,Z}(y, z) = \sum_x p(x, y, z).$$

For all y, z for which $p_{Y,Z}(y, z) > 0$,

$$p_{X|Y,Z}(x|y, z) = \frac{p(x, y, z)}{p_{Y,Z}(y, z)}.$$

Therefore,

$$E(X | Y = y, Z = z) = \sum_x x p_{X|Y,Z}(x|y, z) = \sum_x \frac{x p(x, y, z)}{p_{Y,Z}(y, z)}. \quad (7)$$

This shows that, as a function of Y and Z , at all sample points where $Y = y$ and $Z = z$, $E(X | Y, Z)$ is defined by relation (7).

Similarly, let f be the joint probability density function of three continuous random variables X , Y , and Z . Let $f_{X|Y,Z}(x|y, z)$ be the conditional probability density function of X given that $Y = y$ and $Z = z$. Let $f_{Y,Z}$ be the joint probability density function marginalized over Y and Z ; then

$$f_{Y,Z}(y, z) = \int_{-\infty}^{\infty} f(x, y, z) dx.$$

For all y, z , for which $f_{Y,Z}(y, z) > 0$,

$$f_{X|Y,Z}(x|y, z) = \frac{f(x, y, z)}{f_{Y,Z}(y, z)}.$$

Therefore,

$$E(X | Y = y, Z = z) = \int_{-\infty}^{\infty} x f_{X|Y,Z}(x|y, z) dx = \int_{-\infty}^{\infty} \frac{x f(x, y, z)}{f_{Y,Z}(y, z)} dx. \quad (8)$$

As in discrete case, this shows that, as a function of Y and Z , at all sample points where $Y = y$ and $Z = z$, $E(X | Y, Z)$ is defined by relation (8).

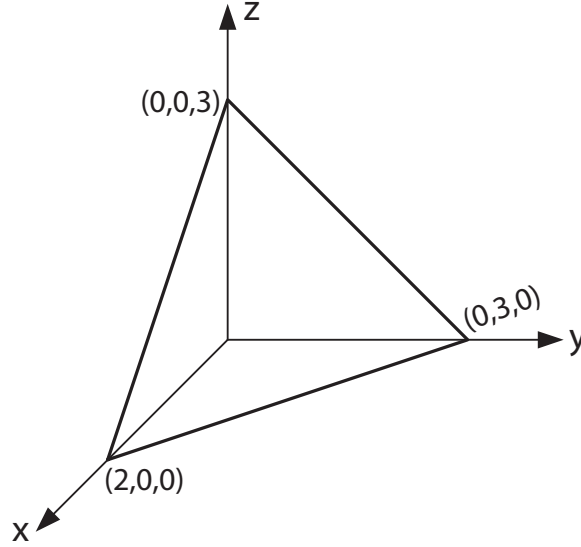


Figure 10d The tetrahedron of Example 10s.

Example 10s Let X , Y , and Z be the coordinates of a point selected randomly from the inside of the tetrahedron of Figure 10d.

- (a) Find $E(Z \mid X, Y)$.
- (b) Verify that $E[E(Z \mid X, Y)] = E(Z)$.

Solution:

- (a) The equation of the plane passing through the points $(2, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 3)$ is $3x + 2y + 2z = 6$. For $0 < x < 2$ and $0 < y < (6 - 3x)/2$, given that $X = x$ and $Y = y$, the random variable Z is uniform over the interval $\left(0, \frac{6 - 3x - 2y}{2}\right)$. Hence

$$E(Z \mid X = x, Y = y) = \frac{6 - 3x - 2y}{4}.$$

This implies that

$$E(Z \mid X, Y) = \frac{6 - 3X - 2Y}{4},$$

where X and Y are jointly distributed random variables and their joint distribution is uniform over the set $\{(x, y) : 0 < x < 2, 0 < y < (6 - 3x)/2\}$.

- (b) The volume of the tetrahedron in Figure 10d is the area of its base, which is $(2 \times 3)/2 = 3$, times one-third of its height, which is 1. So the volume is 3. Let f be the joint probability density function of X , Y , and Z , the coordinates of a point randomly selected from the tetrahedron. Then, by (9.10),

$$f(x, y, z) = \begin{cases} 1/3 & (x, y, z) \in \text{inside the tetrahedron} \\ 0 & \text{otherwise.} \end{cases}$$

Clearly,

$$\begin{aligned} E(Z) &= \int_0^2 \int_0^{(6-3x)/2} \int_0^{(6-3x-2y)/2} \frac{z}{3} dz dy dx = \frac{3}{4}, \\ E(X) &= \int_0^2 \int_0^{(6-3x)/2} \int_0^{(6-3x-2y)/2} \frac{x}{3} dz dy dx = \frac{1}{2}, \\ E(Y) &= \int_0^2 \int_0^{(6-3x)/2} \int_0^{(6-3x-2y)/2} \frac{y}{3} dz dy dx = \frac{3}{4}. \end{aligned}$$

We see that

$$\begin{aligned} E[E(Z | X, Y)] &= E\left(\frac{6 - 3X - 2Y}{4}\right) = \frac{1}{4}[6 - 3E(X) - 2E(Y)] \\ &= \frac{1}{4}[6 - 3(1/2) - 2(3/4)] = \frac{3}{4} = E(Z). \quad \blacklozenge \end{aligned}$$

Example 10t Let the joint probability density function of X , Y , and Z be given by

$$f(x, y, z) = \frac{4}{3}(xy + yz + zx), \quad 0 < x, y, z < 1.$$

Then

$$f_{Y,Z}(y, z) = \int_0^1 \frac{4}{3}(xy + yz + zx) dx = \frac{2}{3}(2yz + y + z).$$

Hence

$$\begin{aligned} E(X | Y = y, Z = z) &= \int_0^1 x f_{X|Y,Z}(x|y, z) dx = \int_0^1 x \frac{f(x, y, z)}{f_{Y,Z}(y, z)} dx \\ &= 2 \int_0^1 \frac{x(xy + yz + zx)}{2yz + y + z} dx = \frac{3yz + 2y + 2z}{3(2yz + y + z)}. \end{aligned}$$

Thus the random variable $E(X | Y, Z)$ is given by

$$E(X | Y, Z) = \frac{3YZ + 2Y + 2Z}{3(2YZ + Y + Z)}. \quad \blacklozenge$$

The following is a natural extension of the definition of $E(X | Y, Z)$.

Definition 10c Let $X, Y_1, \dots, Y_n$ be $n + 1$ random variables. By the random variable $E(X | Y_1, \dots, Y_n)$, we mean a function of the random variables $Y_1, \dots, Y_n$ that is defined to be $E(X | Y_1 = y_1, \dots, Y_n = y_n)$ when $Y_1 = y_1, \dots$, and $Y_n = y_n$.

Let p be the joint probability mass function of the discrete random variables $X, Y_1, Y_2, \dots, Y_n$. Let $p_{X|Y_1, Y_2, \dots, Y_n}(x|y_1, y_2, \dots, y_n)$ be the conditional probability mass function of X given that $Y_1 = y_1, Y_2 = y_2, \dots, Y_n = y_n$. Let $p_{Y_1, Y_2, \dots, Y_n}$ be the joint probability mass function marginalized over $Y_1, Y_2, \dots, Y_n$; that is,

$$p_{Y_1, Y_2, \dots, Y_n}(y_1, y_2, \dots, y_n) = \sum_x p(x, y_1, y_2, \dots, y_n).$$

For all $y_1, y_2, \dots, y_n$ for which $p_{Y_1, Y_2, \dots, Y_n}(y_1, y_2, \dots, y_n) > 0$,

$$p_{X|Y_1, Y_2, \dots, Y_n}(x | y_1, y_2, \dots, y_n) = \frac{p(x, y_1, y_2, \dots, y_n)}{p_{Y_1, Y_2, \dots, Y_n}(y_1, y_2, \dots, y_n)}.$$

Therefore,

$$\begin{aligned} E(X | Y_1 = y_1, Y_2 = y_2, \dots, Y_n = y_n) &= \sum_x x p_{X|Y_1, Y_2, \dots, Y_n}(x|y_1, y_2, \dots, y_n) \\ &= \sum_x \frac{x p(x, y_1, y_2, \dots, y_n)}{p_{Y_1, Y_2, \dots, Y_n}(y_1, y_2, \dots, y_n)}. \end{aligned} \quad (9)$$

This shows that, as a function of $Y_1, Y_2, \dots, Y_n$, at all sample points where $Y_1 = y_1, Y_2 = y_2, \dots$, and $Y_n = y_n$, $E(X | Y_1, Y_2, \dots, Y_n)$ is defined by relation (9).

Similarly, let f be the joint probability density function of the continuous random variables $X, Y_1, Y_2, \dots, Y_n$. Let $f_{X|Y_1, Y_2, \dots, Y_n}(x|y_1, y_2, \dots, y_n)$ be the conditional probability density function of X given that $Y_1 = y_1, Y_2 = y_2, \dots, Y_n = y_n$. Let $f_{Y_1, Y_2, \dots, Y_n}$ be the joint probability density function marginalized over $Y_1, Y_2, \dots$, and Y_n ; then

$$f_{Y_1, Y_2, \dots, Y_n}(y_1, y_2, \dots, y_n) = \int_{-\infty}^{\infty} f(x, y_1, y_2, \dots, y_n) dx.$$

For all $y_1, y_2, \dots, y_n$ for which $f_{Y_1, Y_2, \dots, Y_n}(y_1, y_2, \dots, y_n) > 0$,

$$f_{X|Y_1, Y_2, \dots, Y_n}(x|y_1, y_2, \dots, y_n) = \frac{f(x, y_1, y_2, \dots, y_n)}{f_{Y_1, Y_2, \dots, Y_n}(y_1, y_2, \dots, y_n)}.$$

Therefore,

$$\begin{aligned} E(X | Y_1 = y_1, Y_2 = y_2, \dots, Y_n = y_n) &= \int_{-\infty}^{\infty} x f_{X|Y_1, Y_2, \dots, Y_n}(x|y_1, y_2, \dots, y_n) dx \\ &= \int_{-\infty}^{\infty} \frac{x f(x, y_1, y_2, \dots, y_n)}{f_{Y_1, Y_2, \dots, Y_n}(y_1, y_2, \dots, y_n)} dx. \end{aligned} \quad (10)$$

As in other cases discussed above, this shows that, as a function of $Y_1, Y_2, \dots, Y_n$, at all sample points where $Y_1 = y_1, Y_2 = y_2, \dots$, and $Y_n = y_n$, $E(X | Y_1 = y_1, Y_2 = y_2, \dots, Y_n = y_n)$ is defined by relation (10).

It is important to note that, as in Theorem 10.6 of the book, the expected value of the random variable $E(X | Y_1, Y_2, \dots, Y_n)$ is equal to the expected value of X . To show this

fact, say, for continuous random variables, since $E(X | Y_1 = y_1, Y_2 = y_2, \dots, Y_n = y_n)$ is a function of $Y_1, Y_2, \dots, Y_n$, by Theorem 9.4 of the book,

$$\begin{aligned}
 & E[E(X | Y_1, Y_2, \dots, Y_n)] \\
 &= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} E(X | Y_1 = y_1, \dots, Y_n = y_n) f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) dy_1 \cdots dy_n \\
 &= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} \frac{x f(x, y_1, \dots, y_n)}{f_{Y_1, \dots, Y_n}(y_1, \dots, y_n)} dx \right] f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) dy_1 \cdots dy_n \\
 &= \int_{-\infty}^{\infty} x \left[\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x, y_1, y_2, \dots, y_n) dy_1 dy_2 \cdots dy_n \right] dx \\
 &= \int_{-\infty}^{\infty} x f_X(x) dx = E(X).
 \end{aligned}$$

In particular, this proof implies that

$$E(X) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} E(X | Y_1 = y_1, \dots, Y_n = y_n) f_{Y_1, \dots, Y_n}(y_1, \dots, y_n) dy_1 \cdots dy_n. \quad (11)$$

The proof of $E(X) = E[E(X | Y_1, Y_2, \dots, Y_n)]$ in the discrete case is similar to the continuous case. For discrete random variables, the relation analogous to (11) is

$$E(X) = \sum_{y_1} \cdots \sum_{y_n} E(X | Y_1 = y_1, \dots, Y_n = y_n) P(Y_1 = y_1, \dots, Y_n = y_n). \quad (12)$$

Note that by (11) and (12), $E(X) = E[E(X | Y_1, Y_2, \dots, Y_n)]$ is a condensed way to say that $E(X)$ is the weighted average of the conditional expectations of X (given $Y_1 = y_1, Y_2 = y_2, \dots, Y_n = y_n$) over all possible values of $Y_1, Y_2, \dots$, and Y_n . We have established the following important generalization of Theorem 10.6 of the book, both in the continuous and in the discrete cases.

Theorem 10b Let $X, Y_1, Y_2, \dots, Y_n$ be $n + 1$ random variables. Then

$$E[E(X | Y_1, Y_2, \dots, Y_n)] = E(X).$$

We will now prove two theorems, which in the study of stochastic processes, especially martingales, are proved to be very useful.

Theorem 10c Let $X, Y_1, Y_2, \dots, Y_n$ be $n + 1$ random variables and h be a real valued function from $\mathbf{R}^n$ to $\mathbf{R}$. Then

$$E[h(Y_1, Y_2, \dots, Y_n)X | Y_1, Y_2, \dots, Y_n] = h(Y_1, Y_2, \dots, Y_n)E(X | Y_1, Y_2, \dots, Y_n).$$

Proof: We will show that both the right and left sides are the same for all sample points. Suppose that at a certain sample point we have that $Y_1 = y_1, Y_2 = y_2, \dots$, and $Y_n = y_n$. At that point the right side of the equation is

$$h(y_1, \dots, y_n)E(X | Y_1 = y_1, \dots, Y_n = y_n).$$

The following shows that the left side is also the same.

$$\begin{aligned}
 & E[h(Y_1, \dots, Y_n)X \mid Y_1 = y_1, \dots, Y_n = y_n] \\
 &= E[h(y_1, \dots, y_n)X \mid Y_1 = y_1, \dots, Y_n = y_n] \\
 &= h(y_1, \dots, y_n)E(X \mid Y_1 = y_1, \dots, Y_n = y_n). \quad \blacklozenge
 \end{aligned}$$

Theorem 10d For $n \geq 1$, let $X, W, Y_1, \dots, Y_n$ be $n + 2$ random variables. Then

$$E(W + X \mid Y_1, Y_2, \dots, Y_n) = E(W \mid Y_1, Y_2, \dots, Y_n) + E(X \mid Y_1, Y_2, \dots, Y_n).$$

Proof: At a sample point where $Y_1 = y_1, \dots, Y_n = y_n$, the right side is

$$E(W \mid Y_1 = y_1, \dots, Y_n = y_n) + E(X \mid Y_1 = y_1, \dots, Y_n = y_n).$$

Let f be the joint probability density function of $W, X, Y_1, \dots, Y_n$. We will show that the left side is also the same thing.

$$\begin{aligned}
 & E(W + X \mid Y_1 = y_1, \dots, Y_n = y_n) \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (w + x) \frac{f(w, x, y_1, \dots, y_n)}{f_{Y_1, \dots, Y_n}(y_1, \dots, y_n)} dw dx \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} w \frac{f(w, x, y_1, \dots, y_n)}{f_{Y_1, \dots, Y_n}(y_1, \dots, y_n)} dw dx + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x \frac{f(w, x, y_1, \dots, y_n)}{f_{Y_1, \dots, Y_n}(y_1, \dots, y_n)} dw dx \\
 &= E(W \mid Y_1 = y_1, \dots, Y_n = y_n) + E(X \mid Y_1 = y_1, \dots, Y_n = y_n). \quad \blacklozenge
 \end{aligned}$$

Answers to Exercises of Section 10B

1. (a) 1000; (b) 1,001,000; (c) 1,010,1000.
2. 1066.